

PRANEEL NADGIR

Computer Science Engineer | Multimodal AI & Computer Vision Researcher

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Summary

Computer Science undergraduate and project intern at KLETech-CEVI, specializing in Multimodal AI and Computer Vision. Over the last few years, I have built production-grade deep learning systems focusing on VLMs, diffusion models, and 3D vision. I am incredibly proud to have been named the Global Runner-Up at the CVPR NTIRE 2026 challenge and to have my work published at ICETCI 2026 (Springer Nature). Technically, my foundation is built on PyTorch, CUDA, Docker, and transformer-based architectures.

Education

KLE Technological University, Hubballi
B.E. in Computer Science & Engineering

Expected Aug 2027
CGPA: 9.1/10

Industry Experience

Knit Space (Software Research & Services Pvt. Ltd.)
Software Engineering Intern

Jul 2025 – Aug 2025
Hubballi, Karnataka

- **Tools:** Python, Semantic Retrieval, Docker, Contextual Query Processing, REST APIs.
- Developed a domain-aware semantic retrieval pipeline with contextual indexing, improving query relevance for enterprise-scale backend search workflows.

Publications and Challenges

1. **NTIRE 2026 High-FPS Video Frame Interpolation, CVPR Workshop** [\[GitHub\]](#) [\[Report\]](#)
2. **ADHD Diagnosis & Sex Classification via fMRI Brain Imaging** — WiDS Datathon 2025 [\[Kaggle\]](#)
3. **MK-DCSAU-Net: Segmentation-Guided Melanoma Classification via Multi-Kernel Artifact Removal** — Published at ICETCI 2026 [\[Paper\]](#) [\[Springer Nature\]](#)

Projects and Research

1. **LexiVision: OCR-Augmented Scientific Figure Captioning** Jan 2026 – Present [\[GitHub\]](#)
Python, Qwen2.5-VL-7B, BLIP, FastAPI, React, OCR, CUDA, Docker
 - OCR-augmented VLM pipeline for scientific figure understanding; OCR-injected BLIP achieved **+166.4% BLEU-4** and **+39.7% METEOR** over the FigCaps-HF baseline across **13,355 test samples**. Zero-shot Qwen2.5-VL-7B reached ROUGE-L Recall of **0.3782**, deployed via FastAPI + React web app.
2. **MK-DCSAU-Net: Segmentation-Guided Melanoma Classification** Sep 2025 – Jan 2026 [\[Paper\]](#)
PyTorch, U-Net, EfficientNet-B3, Grad-CAM, CUDA, Medical AI
 - Dual-channel self-attention U-Net with multi-kernel artifact removal for dermoscopic melanoma classification on ISIC datasets. Published at **ICETCI 2026 (Springer Nature)**.
3. **3D Garment Deformation Simulation for Indian Traditional Clothing** May 2026 – Present
Python, PyTorch, ContourCraft, SMPL-X, Blender, CUDA
 - Physically plausible cloth simulation pipeline for Sarees under human body motion on NVIDIA RTX 6000 Ada, extending ContourCraft (SIGGRAPH 2024). First work targeting unstructured drape-based garments absent from existing cloth simulation benchmarks.
4. **3D Human-Object Interaction Synthesis** Sep 2025 – Jan 2026 [\[Paper\]](#)
PyTorch, Diffusion Models, CLIP, Trimesh, SDF, BPS, CUDA
 - Conditional diffusion model for HOI synthesis with zero-shot generalization to novel object categories. Addressed dataset scarcity via proxy-object substitution using custom 3D mesh replacement, validated on constrained compute.

Technical Skills

Languages: Python, C++, Java, C, SQL, MATLAB

Deep Learning: PyTorch, TensorFlow, CUDA, Diffusion Models, VLMs, Transformers, OpenCV, XGBoost

Engineering & Tools: Docker, REST APIs, FastAPI, Git, Linux, Conda, Model Serving, Distributed Training

Interests: AR/VR/XR, MoCap, HoloLens, Meta Quest Development ·

Achievements

NTIRE 2026 Global Runner-Up — Secured 2nd place globally among numerous international teams in the High-FPS VFI Challenge (CVPR Workshop). Achieved 26.316 dB PSNR, trailing 1st place by a narrow 0.02 dB margin.